

Tianyu Cao

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Wuhan, China

github.com/luminescentfrr

RESEARCH INTERESTS

Computational Biology, Generative Modeling, Representation Learning, Biomedical AI

EDUCATION

• Hubei University of Economics

Sep 2023 – Jul 2027 (Expected)

Bachelor of Engineering in Artificial Intelligence

Wuhan, China

- GPA: 3.41/4.00; Major Core GPA: 3.70/4.00
- Honors: Canglong Student Scholarship (2024, 2025 and 2026 Academic Years; awarded to the top 15% of students)

PUBLICATIONS

J = Journal, C = Conference; * Corresponding authors

- [J.1] Tianyu Cao, Mingyao Ma, Haoxiang Zhang, Guoyuan Zhou*, and Ruiheng Li*. **“SegRWKV: Fast and Accurate Biomedical Image Segmentation via a Receptance-Weighted Key-Value Network.”** *Journal of Computational Design and Engineering*, vol. 13, no. 8, pp. 196–211, Aug. 2026. doi:10.1093/jcde/qwag065.
- [C.1] Tianyu Cao, Hongxi Xia, Mingyao Ma, Haoxiang Zhang, Guoyuan Zhou*, and Ruiheng Li*. **“SegDrift: Prior-Guided Semantic Drifting with Hard Negative Mining for Medical Image Segmentation.”** *IEEE International Conference on Bioinformatics & Biomedicine (BIBM 2026)*. Accepted.
- [C.2] Tianyu Cao, Hongxi Xia, Mingyao Ma, Haoxiang Zhang, Xinye Li, Guoyuan Zhou*, and Ruiheng Li*. **“Averaging Models Is Not Averaging Spectra: Steering Federated Representations.”** *The Fifteenth International Conference on Learning Representations (ICLR 2027)*, under review.

RESEARCH EXPERIENCE

• Hubei Key Laboratory of Digital Finance Innovation

Hubei University of Economics; Research Assistant | Advisor: Associate Professor Ruiheng Li

Wuhan, China

• Safety-Constrained Generative Modeling for Partial Cellular Reprogramming

Jul 2026 – Present

Ongoing research on cellular rejuvenation with preservation of cell identity

- Studying partial cellular reprogramming under OSK/OSKM perturbations to balance rejuvenation with cell-identity preservation and biological safety.
- Building a conditional flow-matching framework for continuous cell-state trajectories in multi-omics data and identifying safe reprogramming windows.
- Formulating safety-constrained counterfactual optimization and gene-attribution analyses to generate biologically plausible trajectories while minimizing identity loss and risk.

• SegDrift: Prior-Guided Semantic Drifting for Medical Image Segmentation

May 2026 – Jul 2026



- Identified semantic inconsistencies overlooked by conventional pixel-level losses and introduced a DINOv3-guided semantic-drifting framework with a CLS calibration prior and prediction-aware hard-negative mining.
- Designed a single-step conditional generative process that avoids iterative sampling while preserving semantic consistency; accepted at IEEE BIBM 2026. [code available on GitHub](#)

• Cold-Start Active Learning for Unlabeled Data

Mar 2026 – Present

- Proposed MACS for high-value sample selection in cold-start active learning without labeled data.
- Used C-RADIOv4 to extract semantic, structural, and segmentation-related information in a single forward pass, modeling uncertainty through cross-space inconsistency and multi-resolution features.
- Developed adaptive clustering and non-uniform budget allocation for informative sample selection under limited annotation budgets.

• Federated Learning Theory

Feb 2026 – Jun 2026

- Studied JointCloud federated learning for privacy-preserving, cross-institutional medical image segmentation and identified persistent representation drift under non-IID data despite parameter-level correction.
- Developed a representation-level account of Spectral Representation Collapse using Von Neumann Entropy to quantify residual performance gaps; manuscript submitted to ICLR 2027.

• RWKV-Based Medical Image Segmentation (SegRWKV)

Sep 2025 – Jan 2026



- Proposed SegRWKV, an efficient dense-prediction network with a hybrid CNN–RWKV encoder and lightweight decoder for long-range dependency modeling and feature reconstruction.
- Achieved $4.8\times$ faster inference and $4.5\times$ lower memory usage, with DSC/F1 improvements of 4–22% over CNNs, 8–58% over Transformers, 1–8% over Mamba, and 10–27% over RWKV-UNet.
- Accepted by the *Journal of Computational Design and Engineering*; [code available on GitHub](#).

• Physics-Inspired Medical Image Segmentation (EMG-UNet)

Jun 2025 – Nov 2025



- Proposed a heat-diffusion-inspired segmentation framework with interpretable feature propagation and edge enhancement for anatomical structure and boundary modeling.
- Designed and evaluated an efficient encoder–decoder across multiple medical image segmentation datasets; [code available on GitHub](#).

SELECTED PROJECT

• MultiAgent-Based CodeReviewer

Mar 2026 – Jul 2026



- Built an AI-powered desktop application for multi-file code review and conversational repair using six LangGraph expert agents coordinated for issue deduplication and conflict resolution.
- Integrated Tree-sitter-based project analysis, cross-file symbol search, static analysis, test execution, automated editing, and rollback with a FastAPI/SSE backend and Electron interface; [code available on GitHub](#).

COMPETITIONS

- **Honorable Mention**, Mathematical Contest in Modeling / Interdisciplinary Contest in Modeling (MCM/ICM) May 2025
- **Third Prize (Provincial Level)**, Blue Bridge Cup (Software Competition) May 2025
- **Second Prize (Provincial Level)**, Chinese Collegiate Computing Competition May 2026

TECHNICAL SKILLS

- **Machine Learning:** PyTorch, MONAI, nnU-Net, MMCV, generative modeling, representation learning, OpenCLIP, DINOv3, C-RADIOv4
- **Scientific Computing:** Python, C, NumPy, Pandas, Scanpy, AnnData
- **Systems and Development:** Linux, FastAPI, Electron, LangGraph, Tree-sitter, LaTeX